

Natural Language Processing (NLP) – Module 1

Notes

1. What is NLP?

Natural Language Processing (NLP) is a field of Artificial Intelligence that enables computers to understand,

interpret, and process human language. Humans communicate using languages like English, Hindi, or Kannada,

but computers understand numbers. NLP converts human language into a format that machines can analyze.

Example: "I love Python" → ["I", "love", "Python"]

2. Importance of NLP

Most of the world's data exists in text form such as emails, tweets, reviews, and messages.

Organizations use NLP to analyze and extract useful information from this data.

Examples include analyzing product reviews, language translation, spam detection, and customer support automation.

3. Challenges in NLP

Human language is complex and ambiguous.

Common challenges include:

- Ambiguity – a sentence can have multiple meanings.
- Synonyms – different words can have similar meanings.
- Sarcasm – machines struggle to detect sarcasm.
- Context understanding – words can change meaning depending on context.

Example: "bank" could mean a financial institution or a river bank.

4. Applications of NLP

NLP is widely used in modern technology:

- Chatbots (ChatGPT, customer support bots)
- Machine Translation (Google Translate)
- Sentiment Analysis (analyzing opinions in reviews)
- Spam Detection (Gmail spam filters)
- Voice Assistants (Siri, Alexa, Google Assistant)

5. Text Processing Techniques

Before analyzing text, it must be cleaned and prepared. This process is called text preprocessing.

Typical steps include:

- Converting text to lowercase
- Removing punctuation
- Removing stopwords
- Tokenizing words

Example: "I LOVE Python!!!" → ["love", "python"]

6. Regular Expressions (Regex)

Regular Expressions are patterns used to search and extract information from text.

For example, a regex pattern can identify email addresses in a document.

Example pattern: `\w+@\w+\.\w+`

Example text: "Contact: pavan@gmail.com"

7. Tokenization

Tokenization is the process of splitting text into smaller units called tokens.

Example:

Sentence: "I love Python programming"

Tokens: ["I", "love", "Python", "programming"]

8. Stemming

Stemming reduces words to their root form by removing suffixes.

Examples:

playing → play

running → run

studies → studi

Stemming is fast but sometimes produces incorrect word forms.

9. Lemmatization

Lemmatization converts words into their dictionary form (lemma).

Examples:

running → run

better → good

was → be

It is more accurate than stemming because it uses vocabulary and grammar rules.

10. Part-of-Speech (POS) Tagging

POS tagging identifies the grammatical role of each word in a sentence.

Example:

Sentence: "Pavan runs fast"

Pavan → Noun

runs → Verb

fast → Adverb

This helps machines understand sentence structure.